FED4FED: Statistical Fairness Evaluation Report

Privacy-Preserving Verification for Federated Learning Models

FAIRNESS VERDICT: FAIR (Performance Homogeneous)

Under the Conservative Joint Policy (Disparity detected if either Alg 1 or Alg 2 rejects H0), the global federated model demonstrates statistically fair performance. Neither test found significant evidence to reject the null hypothesis of equal performance at alpha = 0.05. Alg 1 p-value: 0.2950, Alg 2 p-value: 0.3018.

1. Experiment & Architecture Configuration

Experiment ID:	mnist_fair_20261020_170444	**Date / Time:**	2026-10-20 17:17:17
Dataset:	MNIST	**Scenario:**	Fair
Clients (K):	10	**Seed:**	42
FL Rounds:	10	**Final Val Accuracy:**	0.9284
Partition Method:	uniform	**Bootstrap Replications (B):**	200

2. Statistical Fairness Hypothesis Tests

Test Algorithm	Statistic	df	p-value	Significance (α)	Decision
Algorithm 1 (Graybill-Deal / T1)	10.7259	9	0.2950	0.050	**Fail to Reject (Fair)**
Algorithm 2 (Welch-Type F)	1.1840	(9, 810.4)	0.3018	0.050	**Fail to Reject (Fair)**

3. Algorithm 3: Global Performance Confidence Interval

Status: Computed under homogeneous fairness condition.
Confidence Level: 95.0%
Pooled Performance Estimate (Graybill-Deal): 0.9295
Confidence Interval: [0.9245, 0.9345]
Standard Error: 0.0025 (Effective df = 810.4)

4. Sample Decentralized Client Performance Statistics

Client ID	Test Samples (n_i)	Observed Metric	Bootstrap Mean	Variance (S_i^2)	Std Error (SE)
Client 00	1000	0.9210	0.9215	0.000068	0.0083
Client 01	1000	0.9300	0.9300	0.000065	0.0081

Client 02	1000	0.9230	0.9229	0.000081	0.0090
Client 03	1000	0.9400	0.9395	0.000064	0.0080
Client 04	1000	0.9420	0.9426	0.000044	0.0067
Client 05	1000	0.9310	0.9306	0.000066	0.0081
Client 06	1000	0.9330	0.9333	0.000053	0.0073
Client 07	1000	0.9170	0.9172	0.000069	0.0083

Scientific Methodology: Fed4Fed privacy-preserving statistical homogeneity testing using non-parametric bootstrapping, Graybill-Deal common mean estimation, and Welch-type heteroscedastic F-testing. Generated by FED4FED Research Suite v1.0.0.